

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	3	2	1	2	3	2	4	3	3	4	1	2	1	1	2	2	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	3	1	2	3	1	2	1	4	4	3	2	3	4	2	2	2	4	3	1
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	4	1	2	2	4	3	4	1	1	3	4	1	4	2	4	2	1	3	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	4	2	4	3	3	2	2	2	3	1	4	3	4	4	4	1	4	4	3	4
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	2	2	1	3	2	3	4	3	1	3	1	4	4	1	4	4	3	1	3	1
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	1	1	1	4	3	4	2	2	3	3	3	3	4	1	3	3	4	1	1
Q.	121	122	123	124	125	126	127	128	129	130										
A.	2	1	3	4	4	2	3	1	2	1										

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HINT – SHEET

1. Ans (2)

$$a = 2t - 6$$

$$\text{or } \frac{dv}{dt} = 2t - 6$$

$$v = \text{max. when } \frac{dv}{dt} = 0 \text{ i.e. } 2t - 6 = 0 \text{ or } t = 3\text{s}$$

2. Ans (1)

$$a = \frac{120 - 80}{8} = 5 \text{ m/s}^2$$

$$T - 3.5g = 3.5 \times 5$$

$$T = 52.5 \text{ N}$$

3. Ans (3)

$$a_T = \frac{dv}{dt} = 4t + 1.$$

$$a_c = \frac{v^2}{r}$$

$$a = \sqrt{a_c^2 + a_T^2}$$

$$a = 10 \text{ m/s}^2$$

4. **Ans (2)**

Gravitational potential energy of ball gets converted into elastic potential energy of the spring.

$$mg(h + d) = \frac{1}{2} Kd^2$$

$$\text{Net work done} = mg(h + d) - \frac{1}{2} Kd^2 = 0.$$

5. **Ans (1)**

$$\vec{a}_{\text{cm}} = \frac{10 \frac{g}{3} - \frac{5g}{3}}{15} = \frac{5g}{3 \times 15} = \frac{g}{9}$$

6. **Ans (2)**

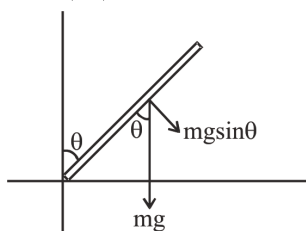
Impulse = Change in momentum

$$= 1(4\hat{i} + \hat{j}) - 1(6\hat{i} + \hat{j}) = -2\hat{i}$$

Which is perpendicular to the wall. Component of initial velocity along $\hat{i} = 6\hat{i}$

$$\Rightarrow \text{Speed of approach} = 6 \text{ m/s and speed of separation} = 4 \text{ m/s Therefore } e = \frac{4}{6} = \frac{2}{3}$$

7. **Ans (3)**

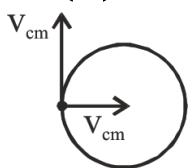


Taking torque about pivot $\tau = I\alpha$

$$mg \sin \theta \frac{l}{2} = \alpha \frac{ml^2}{3}$$

$$\Rightarrow \alpha = \frac{3g}{2l} \sin \theta$$

8. **Ans (2)**



$$v_A = \sqrt{v_{\text{cm}}^2 + v_{\text{cm}}^2} = \sqrt{2} v_{\text{cm}}$$

9. **Ans (4)**

$$Y = \frac{F}{A} \frac{L}{l} = \frac{F d L^2}{M l}$$

As $M = \text{volume} \times \text{density} = A \times L \times d$

$$\therefore A = \frac{M}{Ld}$$

10. **Ans (3)**

where

11. Ans (3)

Given process equation is, $pT^2 = \text{constant}$

Using $PV = nRT$

we can have, $\frac{nRT}{V} \times T^2 = \text{constant}$

or $\frac{T^3}{V} = \text{constant}$

By differentiating above equation on both sides,

$$\frac{3T^2 dT}{V} - \frac{T^3 \times dV}{V^2} = 0$$

$$\text{or } \frac{dV}{V} \times \frac{1}{dT} = \frac{3}{T}$$

In above equation left side term is representing coefficient of volume expansion of gas, as from the definition.

The coefficient of volume expansion is defined as fractional change in volume per unit rise in temperature.

12. Ans (4)

Let the length of the pendulum at 0°C and $t^\circ\text{C}$ is l_0 and l_t respectively.

The time period of oscillations of the pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}} \text{ i.e., } T \propto \sqrt{l}$$

$$\therefore \frac{T_t}{T_0} \sqrt{\frac{l_1}{l_0}} = \sqrt{\frac{l_0(1+\alpha t)}{l_0}} = \sqrt{1+\alpha t}$$

$$= (1+\alpha t)^{1/2} \approx 1 + \frac{1}{2}\alpha t$$

$$\text{or } \frac{T_t}{T_0} - 1 = \frac{1}{2}\alpha t$$

$$\frac{T_t - T_0}{T_0} = \frac{1}{2}\alpha t$$

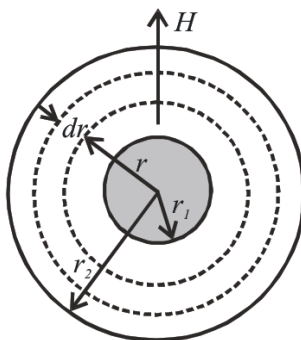
This is the time loss per second.

$$\text{Thus, time loss per day} = \frac{1}{2}\alpha t \times 24 \times 60 \times 60$$

$$= \frac{1}{2}\alpha t \times 86400 \text{ s}$$

13. Ans (1)

Consider a concentric spherical shell of radius r and thickness dr as shown in fig.



The radial rate of flow of heat through this shell in steady state will be

14. **Ans (2)**

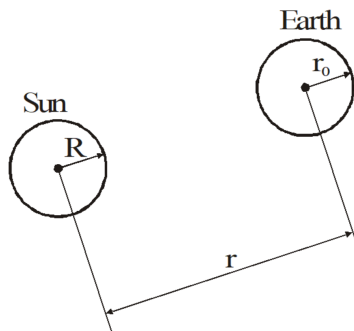
According to Stefan's law

$$\text{Power radiated by Sun} = \sigma(4\pi R^2)T^4$$

Intensity of the sun received by the earth

$$= \frac{\text{Power of the sun}}{4\pi r^2}$$

$$I = \frac{\sigma 4\pi R^2 T^4}{4\pi r^2} = \frac{\sigma R^2 T^4}{r^2}$$



$$\text{Radiant power incident on the earth} = I(\pi r_0^2)$$

where (πr_0^2) is the projection of the earth's area receiving the energy from the sun.

$$P = \frac{\sigma R^2 T^4}{r^2} (\pi r_0^2)$$

15. **Ans (1)**

As forces are weaker, lesser energy is required for expansion

16. **Ans (1)**

$$12.96 \text{ ms}^{-1}$$

$$\text{Heat lost} = \frac{3}{4} \left(\frac{1}{2} m v^2 \right)$$

$$\text{Heat gained} = m(327 - 27) + mL$$

$$\text{As heat lost} = \text{Heat gained}$$

$$\text{So } \frac{3}{4} \left(\frac{1}{2} m v^2 \right) = m \times 300 + m \times L \Rightarrow \frac{3}{8} v^2 = 0.03 \times 1000 \times 4.2 \times 300 + 6 \times 1000 \times 4.2$$

$$\Rightarrow v = 12.96 \text{ m/s}$$

17. **Ans (2)**

$$\Delta U = 2P_0 \Delta V \text{ \& } W = P_0 \Delta V$$

$$\text{so } Q = W + \Delta U = 3P_0 \Delta V$$

$$= 3P_0 V_0$$

18. **Ans (2)**

In same connected liquid, pressure will be same at same horizontal level.

19. **Ans (3)**

$$\frac{KE_1}{KE_2} = \frac{1}{1}$$

20. **Ans (1)**

$$F_{\text{Net}} = -\frac{KQq}{\left(\frac{19d}{2}\right)^2}$$

$$\Rightarrow -\frac{4KQq}{d^2} \frac{1}{361}$$

$$\Rightarrow \frac{qQ}{361 \pi \epsilon_0 d^2}$$

21. **Ans (4)**

Current due to ions having charge $-e$

$$I_1 = n \times e \times v \times A$$

Current due to ions having charge $-2e$

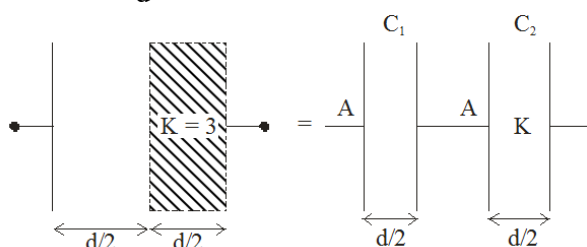
$$I_2 = n \times 2e \times \frac{v}{4} \times A$$

total current $\Rightarrow I_1 + I_2$

$$n \times e \times v \times A \left(1 + \frac{1}{2}\right) = \frac{3}{2} nevA$$

22. **Ans (3)**

$$C_{\text{original}} = \frac{A\epsilon_0}{d}$$



$$C_1 = \frac{A\epsilon_0}{d/2} = \frac{2A\epsilon_0}{d} = C$$

$$C_2 = \frac{KA\epsilon_0}{d/2} = \frac{2KA\epsilon_0}{d} = \frac{6A\epsilon_0}{d} = 3C$$

C_1 & C_2 are in series

$$C_{\text{new}} = \frac{C_1 C_2}{C_1 + C_2} = \frac{C \times 3C}{C + 3C} = \frac{3C}{4}$$

$$= \frac{3}{4} \times \frac{2A\epsilon_0}{d} = \frac{3}{2} \times \frac{A\epsilon_0}{d}$$

$$C_{\text{new}} = \frac{3}{2} C_{\text{original}}$$

$$= \frac{3}{2} \times 4 = 6\mu\text{F}$$

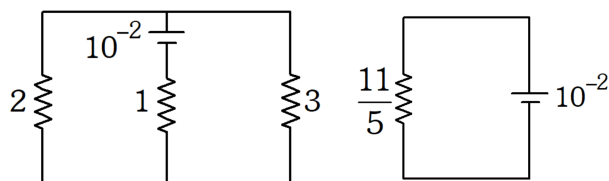
23. **Ans (1)**

Since magnetic force can not change speed of charged particle $\therefore 2 = |\hat{i} + \hat{j} + m\hat{k}|$ or $m = \pm\sqrt{2}$

24. Ans (2)

$$\Sigma = vB\ell = 10^{-2} \text{ V}$$

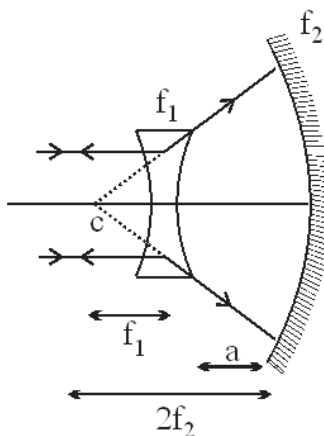
$$I = \frac{10^{-2}}{11/5} = \frac{1}{220} \text{ A}$$



25. Ans (3)

$$f_1 = f_2 = (256) \left(\frac{C}{C-v} \right) \therefore x = y$$

26. Ans (1)



$$2|f_2| - |f_1| = d$$

27. Ans (2)

Let ϕ be the work function of metal.

$$\frac{hc}{\lambda_1} = \phi + \frac{mv_1^2}{2}$$

$$\frac{hc}{\lambda_2} = \phi + \frac{mv_2^2}{2}$$

$$\begin{aligned} hc \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right] &= \frac{m}{2} (v_1^2 - v_2^2) \\ &= \frac{m}{2} [4v_2^2 - v_2^2] \\ &= \frac{3mv_2^2}{2} \end{aligned}$$

$$\frac{mv_2^2}{2} = \frac{hc}{3} \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right]$$

$$\begin{aligned} \phi &= \frac{hc}{\lambda_2} - \frac{mv_2^2}{2} = \frac{hc}{\lambda_2} - \frac{hc}{3} \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right] = \frac{4hc}{3\lambda_2} - \frac{hc}{3\lambda_1} \\ &= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{3} \left[\frac{4}{450 \times 10^{-9}} - \frac{1}{350 \times 10^{-9}} \right] \\ \therefore \phi &= 3.98 \times 10^{-19} \text{ J} \end{aligned}$$

28. **Ans (1)**

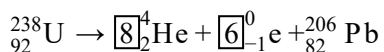
$$mvr = n\hbar$$

$$\frac{mv^2}{r} = kr$$

solving above equation

$$v^2 = \left(\frac{n\hbar}{m} \right) \left(\frac{k}{m} \right)^{1/2} ; r^2 = \left(\frac{n\hbar}{k} \right) \left(\frac{k}{m} \right)^{1/2}$$

29. **Ans (4)**



8 α particles and 6 β particles are emitted.

30. **Ans (4)**

Truth table of the given gate :

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Truth table of option (1)

A	B	C
0	0	1
0	1	1
1	0	0
1	1	1

Truth table of option (2)

A	B	C
0	0	1
0	1	0
1	0	1
1	1	1

Truth table of option (3)

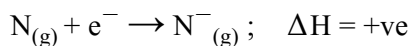
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Truth table of option (4)

A	B	C
0	0	0
0	1	1
1	0	0
1	1	0

Since option (1) has same truth table, hence answer is option (4) is correct.

31. **Ans (3)**



32. **Ans (2)**

The order of radius of 13th group elements is $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$.

Reason $\Rightarrow$ Due to poor shielding effect of d-orbital, radius of Ga is smaller than Al.

33. **Ans (3)**

$$[\text{M}(\text{Co})_x] \text{ EAN} = 36$$

$$36 = 26 - 0 + 2x$$

$$x = 5$$

34. **Ans (4)**

$$(1) \text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3] = 3$$

$$(2) \text{K}_2[\text{Zn}(\text{OH})_4] = 2$$

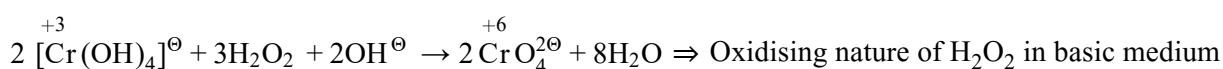
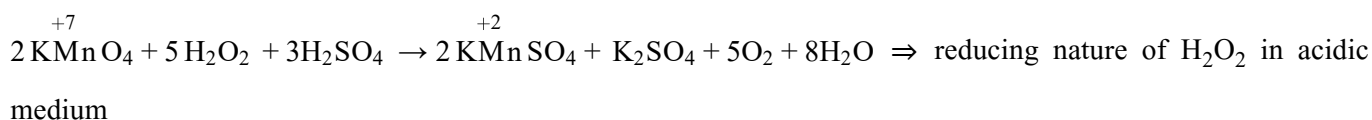
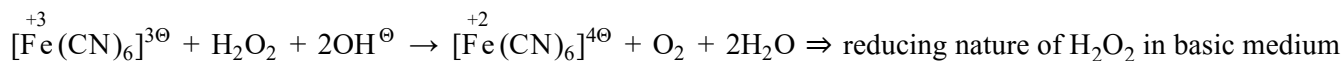
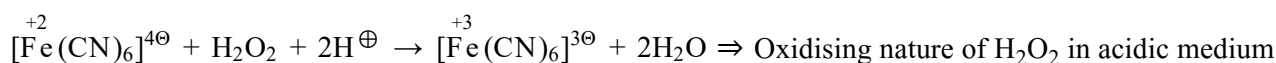
$$(3) \text{K}_2[\text{PtCl}_4] = 2$$

$$(4) \text{K}[\text{Pt}(\text{NH}_3)\text{Cl}_5] = 1$$

35. **Ans (2)**

Temporary hardness of water is due to the presence of $\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$ in water.

36. **Ans (2)**



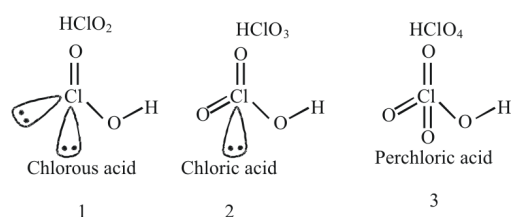
H_2O_2 can act both as oxidising and reducing agent in both acidic and basic Medium.

37. **Ans (2)**

The correct order will be $\text{Xe} > \text{Kr} > \text{Ar} > \text{Ne} > \text{He}$.

38. **Ans (4)**

Number of $\text{Cl} = \text{O}$ bonds



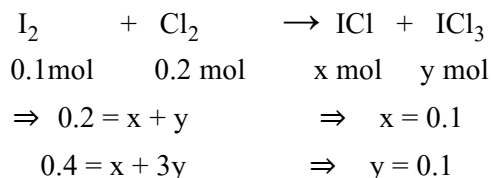
39. **Ans (3)**

$$u = \sqrt{\frac{3RT}{M}} \text{ or } \frac{u_1}{u_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{106.4}{u_2} = \sqrt{\frac{100}{400}} \text{ (Note that pressure does not come in use)}$$

$$\text{or } u_2 = 212.8 \text{ m s}^{-1}$$

40. **Ans (1)**



41. **Ans (3)**

$$h_1 S_1 = h_2 S_2$$

$$h_2 = 3 \text{ cm.}$$

$$P_{\text{Total}} = 79 \text{ cm of Hg}$$

42. **Ans (4)**

$$W = 1.8 \text{ eV}$$

$$E_1 = \frac{1240}{310} = 4 \text{ eV}$$

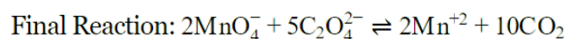
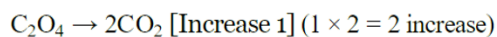
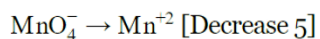
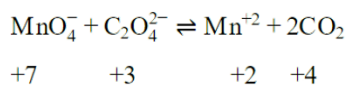
$$E_2 = \frac{1240}{620} = 2 \text{ eV}$$

$$E_3 = \frac{1240}{155} = 8 \text{ eV}$$

$$E_4 = \frac{1240}{1240} = 1 \text{ eV}$$

43. **Ans (1)**

Reaction of oxalate ions being oxidized by MnO_4^- ions is:



Therefore, since 2 moles of MnO_4^- oxidize 5 moles of $\text{C}_2\text{O}_4^{2-}$

So, 1 mole of MnO_4^- will oxidize $\frac{5}{2}$ moles of $\text{C}_2\text{O}_4^{2-}$

44. **Ans (2)**

$$\eta = \frac{T_H - T_C}{T_H} = \frac{500 - 300}{500} = 0.4$$

45. **Ans (2)**

$$\Delta H_{\text{per 1.6g glucose}} = \frac{72 \times 1.6}{180} = 0.64 \text{ Kcal}$$

46. **Ans (4)**

$$\text{SCC} : a = 2R$$

$$\text{BCC} : \sqrt{3}a = 4R$$

$$\text{FCC} : \sqrt{2}a = 4R$$

47. **Ans (3)**

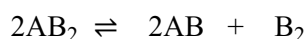
For ideal gas mixture.

$$\Delta H_{\text{mix}} = 0$$

48. **Ans (4)**

$$0 = 0.295 - \frac{0.059}{2} \log K ; \log K = 10 ; K = 10^{10}.$$

49. **Ans (1)**



$$\text{Initial} \quad 2 \quad 0 \quad 0$$

$$\text{Final} \quad 2-2x \quad 2x \quad x$$

$$\text{total no. of moles} = 2 - 2x + 2x + x = 2 + x$$

$$P_{AB_2} = \frac{2-2x}{2+x} \times P, P_{AB} = \frac{2x}{2+x} \times P, P_{B_2} = \frac{x}{2+x} \times P$$

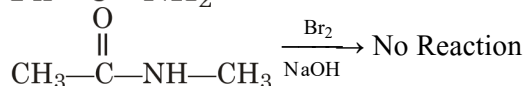
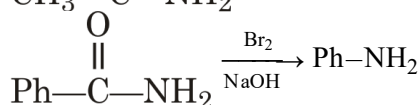
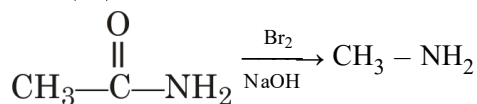
$$K_p = \frac{(P_{AB})^2 (P_{B_2})}{(P_{AB_2})^2} = \frac{\left(\frac{2x}{2+x}\right)^2 P^2 \times \left(\frac{x}{2+x}\right) P}{\left(\frac{2-2x}{2+x}\right)^2 \times P^2} = \frac{x^3 \times P}{(2+x)(1-x)^2}$$

50. **Ans (1)**

$$T_C = \frac{8a}{27Rb}; T_B = \frac{a}{Rb}$$

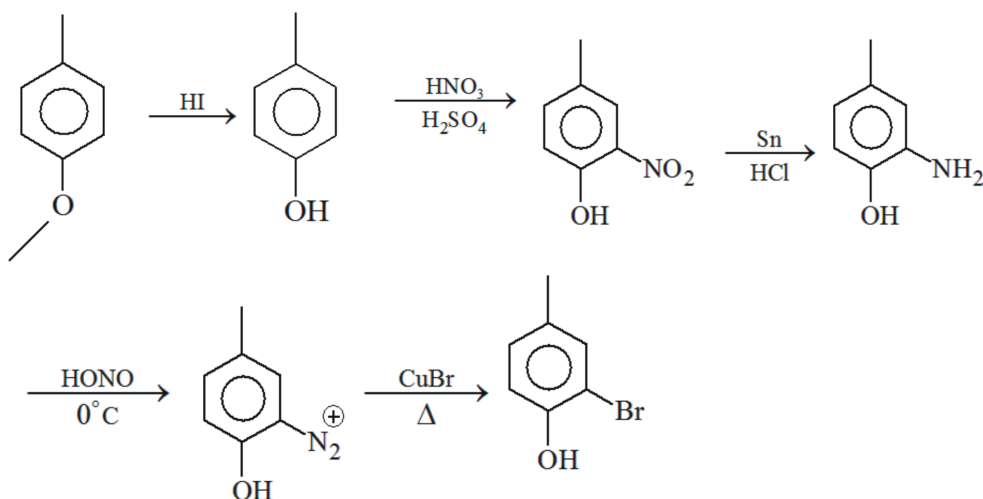
$$T_C = \frac{8T_B}{27}$$

51. **Ans (3)**



(2° amides do not give hoffmann bromamide reaction)

52. Ans (4)



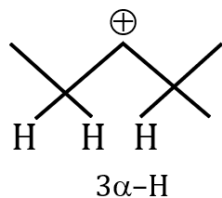
53. Ans (1)

DU = 5

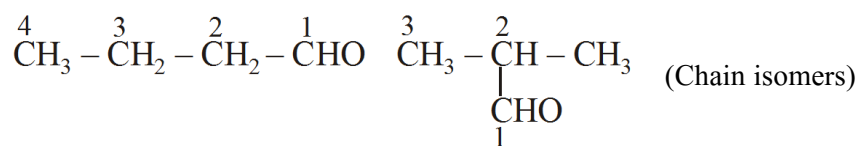
Functional group is Acid halide not ketone.

54. Ans (4)

Hyperconjugating structure $\propto$ Number of α -H



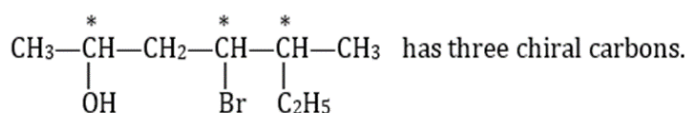
55. Ans (2)



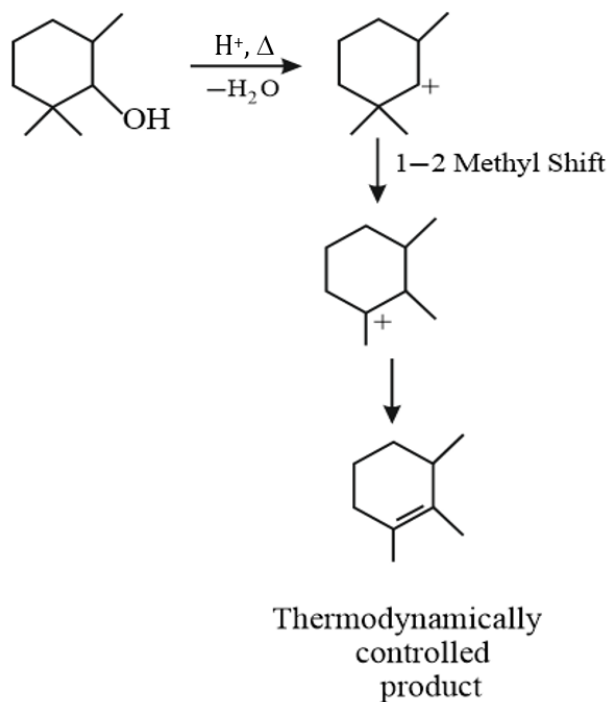
56. Ans (4)

In cycloalkenes G.I. is possible about double bond when there is number of carbon atoms in the ring equal to or greater than 8.

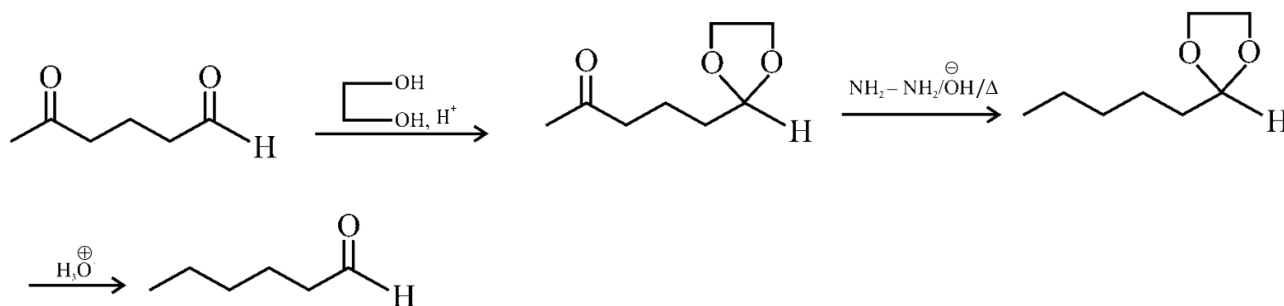
57. Ans (2)



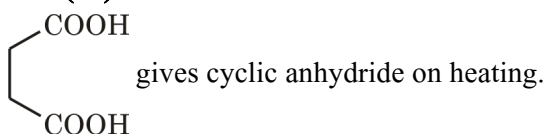
58. Ans (1)



59. Ans (3)



60. Ans (4)



71. Ans (4)

$$\text{As, } \frac{24}{60} = \frac{2}{5}$$

$$\text{Similarly, } \frac{120}{300} = \frac{2}{5}$$

72. Ans (3)

'Konkani' is the language of Goa'. Similarly, Dogri' is the language of 'jammu and Kashmir'.

73. **Ans (4)**

The figures rotate 90° clockwise and heads of the arrow are reversed.

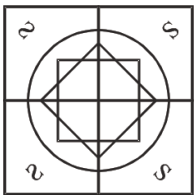
74. **Ans (4)**

By observation,



75. **Ans (4)**

By observation



76. **Ans (1)**

Addition of last two digit is first digit.

77. **Ans (4)**

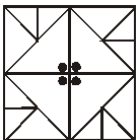
Except Option (4), all follow the same pattern.

78. **Ans (4)**

Only figure (4), when rotates in CW direction does not match with rest of the figure.

79. **Ans (3)**

By observation,



80. **Ans (4)**

By observation

81. **Ans (2)**

In the first row, letters are one step forward i.e. CDE. In the 2nd row, letters are two steps forward i.e. I-K-M

In the third row, the letters are three steps forward i.e., D – – G – – J.

Number is the product of the two numbers.

Hence, $4 \times 7 = 28$

82. **Ans (2)**

First Figure $(6 \times 7) + (8 + 4) = 42 + 12 = 54$

Second Figure $(8 \times 4) + (12 + 7) = 32 + 19 = 51$

Third Figure $(9 \times 5) + (14 + 9) = 45 + 23 = 68$

83. **Ans (1)**

4, 6, 15, 56, 280, 1644

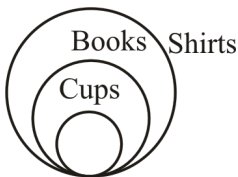
$\rightarrow \times 2 - 2, \times 3 - 3, \times 4 - 4, \times 5 - 5$

84. **Ans (3)**

By observation,

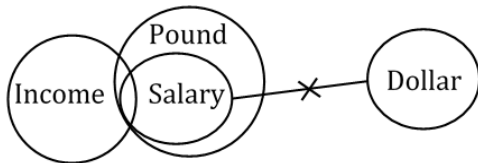


85. **Ans (2)**



Only conclusion 2 follows.

86. **Ans (3)**



87. **Ans (4)**

In every row figures are repeating twice. Hence, [4] follows.



88. **Ans (3)**



89. **Ans (1)**

The given conclusion can be clearly inferred from the line 'The technique, now in India,' which clearly means that the technique was previously not there in India.

90. **Ans (3)**

It is given in the passage that mangoes cultivated in the state are of good quality and the government is trying to export them. This implies the given fact.

91. Ans (1)

$$f'(x) = \begin{cases} -\frac{\pi}{2} \sin \frac{\pi}{2} x & ; \forall 0 \leq x < 1 \\ 5 & ; \forall x \geq 1 \end{cases}$$

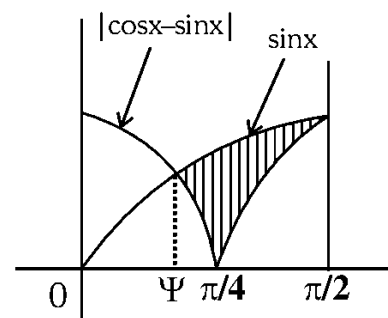
$\Rightarrow f'(x)$ changes its sign from -ive to +ive in the nbd at $x=1$

$\therefore f(x)$ has a local minimum value at $x=1$

92. Ans (4)

$$|\cos x - \sin x| \leq y \leq \sin x \quad \text{Intersection point of } \cos x - \sin x = \sin x \Rightarrow \tan x = \frac{1}{2} \quad \text{Let } \psi = \tan^{-1} \frac{1}{2}$$

$$\text{So, } \tan \psi = \frac{1}{2}, \sin \psi = \frac{1}{\sqrt{5}}, \cos \psi = \frac{2}{\sqrt{5}}$$



$$\begin{aligned} &= \int_{\psi}^{\pi/4} (\sin x - (\cos x - \sin x)) dx \\ &+ \int_{\pi/4}^{\pi/2} (\sin x - (\sin x - \cos x)) dx \\ &= \int_{\psi}^{\pi/4} (2 \sin x - \cos x) dx + \int_{\pi/4}^{\pi/2} \cos x dx \\ &= [-2 \cos x - \sin x]_{\psi}^{\pi/4} + [\sin x]_{\pi/4}^{\pi/2} \end{aligned}$$

$$\begin{aligned} &= -\sqrt{2} - \frac{1}{\sqrt{2}} + 2 \cos \psi + \sin \psi + \left(1 - \frac{1}{\sqrt{2}}\right) \\ &= -\sqrt{2} - \frac{1}{\sqrt{2}} + 2 \left(\frac{2}{\sqrt{5}}\right) + \left(\frac{1}{\sqrt{5}}\right) + 1 - \frac{1}{\sqrt{2}} \end{aligned}$$

$$\text{Area} = \int_{\psi}^{\pi/2} (\sin x - |\cos x - \sin x|) dx = \sqrt{5} - 2\sqrt{2} + 1$$

93. Ans (4)

$$\log_3 x [2 + 4 + 6 + \dots + 16] = 36$$

$$\log_3 x \left[8 \times \frac{18}{2} \right] = 36$$

$$\log_3 x = \frac{1}{2}$$

$$x = \sqrt{3}$$

94. Ans (1)

$${}^{10}C_r \frac{x^{10-r}}{2^{10-r}} \left(\frac{-3}{x^2} \right)^r$$

$$= {}^{10}C_r x^{10-r-2r} (-3)^r \times (2)^{r-10}$$

$$\text{Now } 10 - 3r = 4$$

$$\boxed{r = 2}$$

$$\therefore \text{Coeff of } x^4 = {}^{10}C_2 9 \times \frac{1}{256}$$

95. Ans (4)

$$f(x) = x^{3/2} - \sqrt{x^3 + x^2}$$

$f(x)$ is differentiable & continuous at $x = 2, 5, 1$

$$\text{at } x = 0 \Rightarrow f(0) = f(0+h) = f(0-h)$$

$$f'(0^+) = \text{R.H.D.} = \frac{f(0+h) - f(0)}{h} = \sqrt{h} - \sqrt{h+1}$$

$$= -1$$

96. Ans (4)

If $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other, then $(\sin x - i \cos 2x) = \cos x - i \sin 2x$

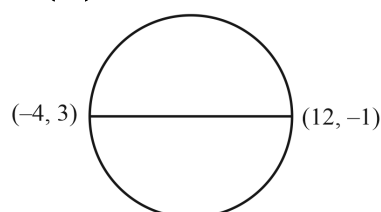
$$\Rightarrow \sin x - i \cos 2x = \cos x - i \sin 2x$$

$$\Rightarrow \sin x = \cos x \text{ and } \cos 2x = \sin 2x$$

$$\Rightarrow \tan x = 1 \text{ and } \tan 2x = 1$$

Both these equations cannot hold simultaneously. Hence the correct answer is (4).

97. Ans (3)



$$\text{Circle: } (x+4)(x-12) + (y-3)(y+1) = 0$$

$$x^2 + y^2 - 8x - 2y - 51 = 0$$

$$y - \text{intercept} = 2\sqrt{f^2 - c}$$

$$= 2\sqrt{1^2 - (-51)} = 2\sqrt{52}$$

$$= 4\sqrt{13}$$

98. Ans (1)

The given D.E.

$$e^{xy}(x dy + y dx) = e^{\frac{x}{y}} \left(\frac{y dx - x dy}{y^2} \right)$$

$$\Rightarrow e^{xy} \cdot d(xy) = e^{\frac{x}{y}} d\left(\frac{x}{y}\right)$$

$$\Rightarrow xy = \ln(e^{x/y} + c)$$

99. Ans (3)

$$(a^2 - 2a - 15)e^{ax} + (b^2 - 2b - 15)e^{bx} = 0$$

$$\Rightarrow (a^2 - 2a - 15) = 0 \text{ and } b^2 - 2b - 15 = 0$$

$$\Rightarrow (a - 5)(a + 3) = 0 \text{ and } (b - 5)(b + 3) = 0$$

$$\Rightarrow a = 5 \text{ or } -3 \text{ and } b = 5 \text{ or } -3$$

$$\Rightarrow a \neq b \text{ hence } a = 5 \text{ and } b = -3$$

$$\text{or } a = -3 \text{ and } b = 5$$

$$\Rightarrow ab = -15$$

100. Ans (1)

$$\text{Put } x = 2a \sin^2 \theta$$

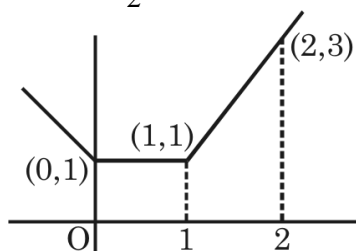
$$dx = 4a \sin \theta \cos \theta d\theta$$

$$I = \int_0^{\pi/2} (2a)^{\frac{5}{2}} \sin^5 \theta (2a)^{-\frac{1}{2}} (\cos \theta)^{-1} 4a \sin \theta \cos \theta d\theta$$

$$(2a)^2 4a \int_0^{\pi/2} \sin^6 \theta d\theta = 16a^3 \frac{5.3.1}{6.4.2} \cdot \frac{\pi}{2} = \frac{5}{2} \pi a^3$$

101. Ans (3)

$$I = 1.1 + \frac{1}{2}(1+3).1 = 3$$



102. Ans (1)

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 6 \neq 0 \text{ consistent}$$

$\therefore$ Homogenous equation

$$\Rightarrow x = y = z = 0$$

$\Rightarrow$ Trivial solution.

103. Ans (1)

$$\Delta(0) = \begin{vmatrix} 3 & -5 & 3 \\ 4 & 1 & 9 \\ 9 & -6 & 21 \end{vmatrix} = 3(21 + 54) + 3(84 - 81) + 3(-24 - 9) = 75 \times 3 + 15 - 99$$

$$d = 225 + 15 - 99 = 240 - 99 = 141$$

$$\Delta(1) = \begin{vmatrix} -1 & -3 & 3 \\ 8 & 7 & 9 \\ 10 & 8 & 21 \end{vmatrix} = a + b + c + d = \begin{vmatrix} -1 & -3 & 3 \\ 8 & 7 & 9 \\ 2 & 1 & 12 \end{vmatrix}$$

$$= -(75) + 3(96 - 18) + 3(8 - 14) = -75 + 78 \times 3 - 18$$

$$= -93 + 234 = 141$$

$$a + b + c + d = 141 \Rightarrow a + b + c = 0$$

104. Ans (1)

$$\frac{2b^2}{a} = 5 \dots\dots(1) \text{ and } 1 - \frac{b^2}{a^2} = \frac{4}{9} \dots\dots(2)$$

$$\text{Solving (1) and (2) gives } a^2 = \frac{81}{4} \text{ and } b^2 = \frac{45}{4}$$

105. Ans (4)

$$f(x) = \frac{x^2}{2} + \cos x + 1, x \geq 0$$

$$\Rightarrow f'(x) = x - \sin x \Rightarrow 0 \Rightarrow f(x) \text{ is inc}$$

$$f(x) = -\frac{x^2}{2} + \cos x + 1, x < 0$$

$$\Rightarrow f'(x) = -x - \sin x > 0$$

$$\Rightarrow f(x) \text{ is inc.}$$

106. Ans (3)

$$f(x) = \sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right]$$

$$\text{Domain : } -1 \leq \left[x^2 - \frac{1}{2} \right] \leq 1 \Rightarrow x \in \left(-\sqrt{\frac{5}{2}}, \sqrt{\frac{5}{2}} \right)$$

$$\text{and } -1 \leq \left[x^2 + \frac{1}{2} \right] \leq 1 \Rightarrow x \in \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}} \right)$$

$$\Rightarrow \text{domain is } x \in \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}} \right) \text{ or } x^2 \in \left[0, \frac{3}{2} \right)$$

$$\text{if (i) } x^2 \in \left[0, \frac{1}{2} \right), \text{ then } f(x) = \pi$$

$$\text{if (ii) } x^2 \in \left[\frac{1}{2}, 1 \right), \text{ then } f(x) = \pi$$

$$\text{if (iii) } x^2 \in \left[1, \frac{3}{2} \right), \text{ then } f(x) = \pi$$

$$\Rightarrow \text{range} = \{\pi\}$$

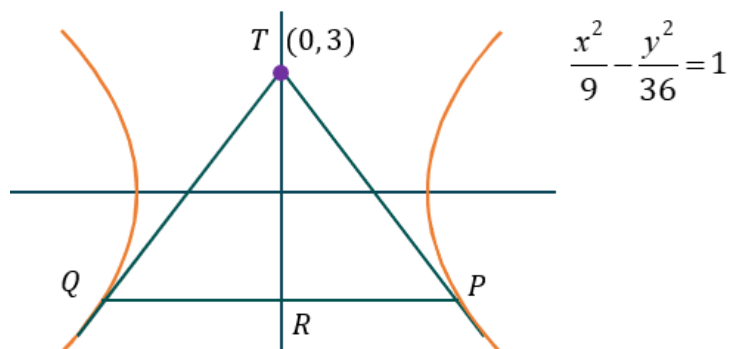
107. Ans (4)

Ans. (4)

Equation PQ : chord of contact $T = 0$

$$y = -12$$

$$\text{Area} : \frac{1}{2} PQ \cdot TR$$



$$TR = 3 + 12 = 15,$$

Solving $y = -12$ and the hyperbola we get Point $P(3\sqrt{5}, -12)$ & $Q(-3\sqrt{5}, -12)$

$$\Rightarrow PQ = 6\sqrt{5}$$

$$\text{Area of } \triangle PTQ = \frac{1}{2} \cdot 15 \cdot 6\sqrt{5} = 45\sqrt{5} \text{ sq. units}$$

108. Ans (2)

Given integral, $I = \int \sin(\cos x) \cdot \sin x \cdot \cos(\cos x) dx$

Let, $\sin(\cos x) = t$

$$\Rightarrow -\cos(\cos x) \cdot \sin x \, dx = dt$$

$$\therefore I = \int -t dt = -\frac{t^2}{2} + C$$

$$= C - \frac{\sin^2(\cos x)}{2}$$

109. Ans (2)

$$f(x) = 1 - x - x^3; f'(x) = -1 - 3x^2 < 0$$

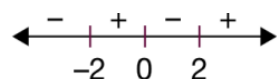
So $f(x)$ is strictly decreasing.

$$\text{Given inequality } 1 - f(x) - f^3(x) > f(1 - 5x)$$

$$\Rightarrow f(f(x)) > f(1 - 5x) \Rightarrow f(x) < 1 - 5x$$

$$\Rightarrow 1 - x - x^3 < 1 - 5x$$

$$\Rightarrow x^3 - 4x > 0 \Rightarrow x(x - 2)(x + 2) > 0$$



Hence, $x \in (-2, 0) \cup (2, \infty)$

110. **Ans (3)**

$$T_1 = \sin^{-1} \left(\frac{1}{\sqrt{10}} \right) = \tan^{-1} \left(\frac{1}{3} \right) = \tan^{-1} 2 - \tan^{-1} 1$$

$$T_2 = \sin^{-1} \left(\frac{1}{\sqrt{50}} \right) = \tan^{-1} \left(\frac{1}{7} \right) = \tan^{-1} 3 - \tan^{-1} 2$$

$$T_{10} = \tan^{-1} 101 - \tan^{-1} 100$$

$$\text{Sum} = \tan^{-1} 101 - \tan^{-1} 1 = \tan^{-1} \left(\frac{100}{102} \right) = \tan^{-1} \left(\frac{50}{51} \right)$$

$$p + q = 50 + 51 = 101$$

111. **Ans (3)**

$$3^{100} \left(\frac{1}{3^{99x}} + \frac{1}{3^{98x}} + \dots + 1 \right)^{1/x}$$

$$L = 3^{100}$$

$$\log_3 L = 100$$

112. **Ans (3)**

Actual data is more close to mean, therefore less variance.

113. **Ans (3)**

$\sim (p \vee q) = \sim p \wedge \sim q$ = Sachin Tendulkar is not a good cricketer and Mukesh Ambani is not a rich person in India So statement-1 is correct and statement 2 is false.

114. **Ans (4)**

$$(A + B)^2 = A + B$$

$$AB = -BA$$

$$\Rightarrow ABA = -BA^2$$

$$-AAB = -BA$$

$$= -A^2B = AB$$

$$= -AB = AB$$

$$\Rightarrow (AB = 0)$$

115. **Ans (1)**

$$\text{As } A^2 = O, A^k = O \forall k \geq 2$$

$$\text{Thus, } (A + I)^{50} = I + 50A$$

$$\Rightarrow (A + I)^{50} = I + 50A$$

$$\Rightarrow a = 1, b = 0, c = 0, d = 1$$

116. **Ans (3)**

Probability of the card being a spade or an ace]

$$= \frac{16}{52} = \frac{4}{13}. \text{ Hence odds in against is } 9 : 4$$

117. Ans (3)

For a particular house being selected,

$$\text{Probability} = \frac{1}{3}$$

$$\text{Probability (all the persons apply for the same house)} = \left(\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} \right) 3 = \frac{1}{9}$$

118. Ans (4)

–N–I–N–T–N– (Gat method)

$$\text{forN} \leftarrow \frac{5!}{3!} \times 6_{C_3} \times 1 \xrightarrow{\text{fore}} = \frac{5!}{3!} \times \frac{6_{P_3}}{3!}$$

119. Ans (1)

$$4 \quad 6 \quad 6 \quad 6 \quad 6 \quad 6 = 4 \times 6^5 = 31104$$

← No of ways →

$$\text{but } 31104 - 1 = 31103 \text{ (Excluding 600000)}$$

120. Ans (1)

Clearly, directrix is $y = a$.

$$SA = AZ$$

Let $P(x, y)$ any point on the parabola.

$$\text{So } PS = PM$$

where PM is $\perp$ from P upon the directrix.

$$\sqrt{(x-0)^2 + (y-3)^2} = |y-9|$$

$$x^2 + (y^2 - 6y + 9) = (y-9)^2 = y^2 - 18y + 81$$

$$x^2 + 12y = 72$$

121. Ans (2)

$$\left. \begin{array}{l} x = t^2 + 1 \Rightarrow t^2 = x - 1 \\ y = 2t + 1 \Rightarrow t = \frac{y-1}{2} \end{array} \right\} \begin{array}{l} (y-1)^2 = 4(x-1) \\ v(1, 1) \end{array}$$

122. Ans (1)

$$x^2 + px + 12 = 0$$

$$\text{put } x = 4$$

$$16 + 4p + 12 = 0$$

$$\Rightarrow p = -7$$

$$\text{Now } x^2 + px + q = 0$$

$$\Rightarrow x^2 - 7x + q = 0$$

This quadratic equation has equal root

$$\therefore D = 0$$

$$\Rightarrow (-7)^2 - 4(1)(q) = 0$$

$$\Rightarrow 4q = 49$$

$$\Rightarrow q = \frac{49}{4}$$

$$p + q = \frac{-28 + 49}{4} = \frac{21}{4}$$

123. Ans (3)

Let the line is $3x + y = K$

$\therefore$ it passes through $(2, 2) \Rightarrow 6 + 2 = K \Rightarrow K = 8$

$$\Rightarrow \text{line is } 3x + y = 8 \Rightarrow \frac{x}{8/3} + \frac{y}{8} = 1$$

$$\Rightarrow \text{sum of intercepts} = \frac{8}{3} + 8 = \frac{32}{3}$$

124. Ans (4)

By using venn diagram $A - (B \cup C)$

125. Ans (4)

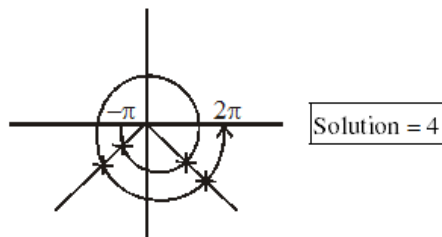
$$2x^3 - 5x^2 - 7x + 3 = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$$

$$\text{H.M.} = \frac{3}{\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}} = \frac{3\alpha\beta\gamma}{\alpha\beta + \beta\gamma + \gamma\alpha} = \frac{3\left(-\frac{3}{2}\right)}{\left(-\frac{7}{2}\right)} = \frac{9}{7}$$

126. Ans (2)

Mid-point z coordinate = 0

127. Ans (3)



128. Ans (1)

$$\cos^2\left(\frac{\pi}{3} - x\right) - \cos^2\left(\frac{\pi}{3} + x\right)$$

$$= \sin\left(\frac{2\pi}{3}\right) \sin(-2x)$$

$$= -\frac{\sqrt{3}}{2} \times \sin x$$

$$= -\frac{\sqrt{3}}{2} \times \sin 2x$$

$$\max \frac{\sqrt{3}}{2}$$

129. Ans (2)

$$\text{Since, } (\vec{a} + 2\vec{b}) \cdot (5\vec{a} - 4\vec{b}) = 0$$

$$\Rightarrow 5|\vec{a}|^2 + 6\vec{a} \cdot \vec{b} - 8|\vec{b}|^2 = 0$$

$$\Rightarrow 6\vec{a} \cdot \vec{b} = 3 \quad \left[\because |\vec{a}| = |\vec{b}| = 1 \right]$$

$$\Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

130. Ans (1)

$$\vec{V} = (2\hat{i} - 3\hat{j} + \hat{k}) \times (\hat{i} + 2\hat{j} - 5\hat{k})$$